

Retell AI Topic 5

Webhooks & Call Lifecycle Events

LMS Assessment Documentation

Course: Retell AI – Hands-On Beginner Course

Topic: 5 – Webhooks & Call Lifecycle Events

1. Assessment Overview

This assessment evaluates the ability to configure Retell AI webhooks, capture call lifecycle events, validate webhook payloads, and extract transcripts, recording URLs, and call metadata.

2. Assessment Objective

Demonstrate the ability to configure secure webhook integrations and successfully process call event payloads.

3. Prerequisites

- Review the Requirement Document.
- Review the Implementation Guide.
- Complete Topics 1 through 4.
- Active Retell AI account and access to a webhook inspection tool.

4. Assessment Tasks

- Exercise 1 – Register a Webhook Endpoint
- Exercise 2 – Subscribe to Call Lifecycle Events
- Exercise 3 – Configure Webhook Security
- Exercise 4 – Trigger a Test Call
- Exercise 5 – Capture and Validate Webhook Payloads
- Exercise 6 – Extract Transcript, Recording URL, Duration, and Cost
- Exercise 7 – Milestone Project: Hook into Call Ending and Fetch Recordings

5. Practical Assessment Checklist

- ■ Webhook Endpoint Registered
- ■ Call Started and Call Ended Events Enabled
- ■ Signing Secret Generated
- ■ Webhook Payload Received
- ■ HTTP Response Successful
- ■ Transcript Extracted
- ■ Recording URL Identified
- ■ Call Duration Verified
- ■ Call Cost Verified
- ■ JSON Payload Validated
- ■ No Webhook Processing Errors

6. Assessment Success Criteria

- Configure webhook endpoint successfully.
- Receive lifecycle event payloads.
- Validate webhook configuration and payload delivery.
- Extract transcript and recording URL correctly.
- Process call metadata without errors.

7. Required Deliverables

- Loom/YouTube Demo Video (Mandatory)
- Webhook Configuration Screenshots
- Webhook Payload JSON Evidence
- Transcript and Recording URL Evidence
- Documentation (PDF/DOCX)
- Source Code / GitHub Repository (if applicable)
- Additional Comments (Recommended)

8. Submission Guidelines

Submit the completed assessment through the LMS with the demo video, webhook configuration evidence, payload screenshots, documentation, source code repository where applicable, and implementation notes.

9. Final Submission Checklist

- ■ Completed all assessment exercises
- ■ Verified webhook events
- ■ Uploaded demonstration video
- ■ Attached payload screenshots and evidence
- ■ Reviewed all deliverables

10. Important Note

Before submitting, ensure all required evidence is accessible. Missing demonstration videos, webhook evidence, or required deliverables may result in an incomplete evaluation.

Requirement & Implementation Summary

Learning scope includes registering HTTPS webhook endpoints, subscribing to call_started, call_ended, and call_analyzed, inspecting JSON payloads, extracting call metadata, understanding webhook signature security, and using Webhook.site for real-time inspection.

Implementation Steps

- Open <https://webhook.site/> and copy the unique HTTPS URL.
- Retell Dashboard → Settings → Webhooks; paste the Webhook.site URL.
- Use the Retell webhook Test function to validate endpoint delivery.
- Open Agents → Trainee_Sarah_Receptionist → Test Audio / Playground.
- Run a short web test call and end it.
- Inspect Webhook.site for call_started and subsequent lifecycle payloads.
- Verify transcript, recording URL, duration, and cost fields when present.

Technical Requirements

- Webhook requests use HTTPS POST with application/json.
- Webhook requests include Retell signature information for authenticity verification.
- Payloads contain call metadata and lifecycle event information.
- Webhook delivery is asynchronous.

Test Cases

- TC-001: Register Webhook Endpoint
- TC-002: Configure Event Subscriptions
- TC-003: Cryptographic Signature Verification Analysis
- TC-004: Trigger Webhook via Playground Call
- TC-005: Verify Call Start Payload
- TC-006: Verify Call End / Call Analyzed Payload

Observed Evidence

- Webhook.site endpoint: <https://webhook.site/5f95e2c4-87d3-4114-bf4a-5d9bdb540666>
- Retell Settings → Webhooks configured with the endpoint.
- Webhook Test delivered a POST request to Webhook.site.
- Observed event: call_started.
- Observed call_type: web_call.
- Observed call_status: ongoing.
- Observed Content-Type: application/json.
- Observed x-retell-signature header.
- Observed test call cost: combined_cost = 0 in the start payload.

Important Scope Note

This Topic 5 implementation uses the Retell Playground/Test Audio web-call flow. A phone number purchase or inbound PSTN call is not required for this Topic 5 webhook demonstration.

Submission Links

Loom: <https://www.loom.com/share/08b977186d4b416b85740f966de3d9a7>

GitHub: <https://github.com/shaikshahid777/retell-ai-topic-5-webhooks-call-lifecycle-events>